MIL-S-45428D (MU)
7 May 1969
Superseding
MIL-S-45428C (MU)
30 November 1964

MILITARY SPECIFICATION

SQUIB, ELECTRIC, M2,
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing for one type of electrical initiation device designated as Squib, Electric, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8799892 - Squib, Electric, M2, Assembly
8796522 - Marking Diagram and Sealing for Wood Packing Boxes

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8799892 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C., 20402. Orders shall specify 49 CFR 0-190 (latest revision).)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.- The squib assembly shall comply with all requirements specified on Drawing (dwg.) 8799892 and with all requirements specified in applicable specifications.

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3.3 Volatile content of flash charge.- The total volatile content of the flash charge shall not exceed eight tenths (0.80) percent when determined as specified in 4.4.1.

3.4 Non-functioning.- The squib assembly shall not function when a direct current of two hundred (200) milliamperes is applied for a minimum (min.) of five (5) seconds. The test shall be performed as specified in 4.4.2.

3.5 Functioning.- The squib assembly shall function in an elapsed time not exceeding one tenth (.10) second when tested as specified in 4.4.3.

3.6 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.7 First article inspection.- This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings on the volatile content of the flash charge and the flash charge ingredients shall accompany each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of thirty-two (32) squib assemblies without any flash charge which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including a certificate of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or

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portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period of time in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- The squib assemblies without flash charge will be subjected by the Government to any or all of the applicable examinations or tests specified in 4.3.2 and 4.3.3 of this specification.

4.2.3 Rejection.- If any assembly fails to comply with any or all of the specified requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly in the sample to comply with any or all of the requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article sample inspection. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in

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accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of electric squibs shall contain:

a. Parts from not more than one supplier.

b. Flash charge from not more than one batch or blend of batches which results in homogenity of all flash charge used in the lot.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Sleeve (see dwg. 8836064 covering a detail of dwg. 8799892).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major	AQL 0.40 percent		
101.	Outside diameter.....	Gage	01001
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	01002

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4.3.2.2 Holder assembly (see dwg. 8836067 covering a detail of dwg. 8799892).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 1.00 percent		
101.	Outside diameter of holder.....	Gage	02001
102.	Length of wire protrusion at bridge end.....	Gage	02002
103.	Length of insulated wire protrusion, min.....	Gage	02003
104.	Length of wire with insulation removed, min.....	Gage	02004
105.	Insulation above flush at bridge end of holder.....	Visual	02005
Minor	AQL 1.00 percent		
201.	Cavity or indention over maximum (max.).....	Scale	02006
202.	Evidence of poor workmanship (see 3.6).....	Visual	02007

4.3.2.3 Squib assembly, prior to coating bridge wire with flash charge composition (see dwg. 8799892).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.65 percent		
101.	Bridge wire tension not relieved..	Visual	03001
102.	Plating flaking, peeling, or blistering (when applicable).....	Visual	03002
103.	Presence of soldering flux (when applicable).....	Visual	03003
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	03004

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4.3.2.4 Squib assembly, prior to loading flash charge composition (see dwg. 8799892).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.40 percent		
101.	Sleeve loose	Manual	04001
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	04002

4.3.2.5 Squib assembly (see dwg. 8799892).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Lead wires not shorted	Visual	05001
Major	AQL 0.65 percent		
101.	Length of holder and sleeve, max.	Gage	05002
102.	Charge incomplete or above flush.	Visual	05003
103.	Length of waterproof coating, max.	Scale	05004
104.	Bare wire protrudes beyond surface of holder	Visual	05005
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	05006

4.3.3 Testing.- Unless otherwise specified herein, the following tests shall be performed using MIL-STD-105 for sampling plan requirements.

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4.3.3.1 Volatile content of flash charge (see 3.3)- Major defect - Code No. 06001. - The contractor shall provide adequate controls to insure that the flash charge complies with the requirement. For verification, the contractor shall select a sample for test from that quantity of flash charge used in each eight hours of loading production. A composite sample shall not be used. If the volatile content of a sample exceeds the requirement and loading has not begun, that quantity of flash charge represented by the sample shall be rejected. If components have been loaded with flash charge containing excessive volatiles, the remaining unloaded flash charge and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1.

4.3.3.2 Non-functioning (see 3.4) - Critical defect - Code No. 05007.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.2.

4.3.3.3 Functioning (see 3.5 and Table I)

4.3.3.3.1 Sampling.- One hundred and forty-five (145) squib assemblies shall be selected from each lot for test. If a retest is necessary two hundred and eighty (280) additional squib assembly samples shall be selected from the lot. The number of samples to be subjected to each test shall be as specified in Table I.

TABLE I

Test	Classification	Code No.	No. of Samples 1st	No. of Samples Retest	Test Paragraph
Water immersion	Major	05008	70*	140	4.4.3.1
Cold temperature	Major	05009	70*	140	4.4.3.2
Duds (see 6.4)	Major	05010	-	-	-

*Duds shall be replaced from the five (5) extra samples selected.

4.3.3.3.2 Rejection.- The lot shall be rejected if any of the following test results occur:

a. Water immersion.- The mean functioning time ($\bar{X}$) plus the product of 2.42 and the standard deviation of the sample exceeds .10 second (see 6.5). Duds shall not be used in the calculations.

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b. Cold temperature.- The mean functioning time ($\bar{X}$) plus the product of 2.42 and the standard deviation of the sample exceeds .10 second. Duds shall not be used in the calculation.

c. Duds.- If, during the combined water immersion and cold temperature tests, two (2) or more duds are found, the lot shall be rejected. If only one dud is found a retest is permitted using the sample of two hundred and eighty (280) squib assemblies. If one more dud is found during the retest, the lot shall be rejected.

4.3.3.4 Lead wire pull test (see dwg. 8836067) - Major defect - Code No. 02008.- The sampling plan for this test shall be in accordance with Code Letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.

4.3.3.5 Lead wire and plug leakage test (see dwg. 8836067) - Major defect - Code No. 02009.- The sampling plan for this test shall be in accordance with Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.5.

4.3.3.6 Bridge wire load test (see dwg. 8799892) - Major defect - Code No. 03005.- The sampling plan for this test shall be in accordance with Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.6.

4.3.3.7 Bridge wire resistance (see dwg. 8799892) - Major defect - Code No. 05011.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.7. Any assembly which fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.8 Determination of flash charge composition (see dwg. 8799892) - Major defect - Code No. 07001.- A sample of sufficient size to perform the composition determination shall be selected from each batch or blend of batches used in loading the squib assembly lot. If the composition fails to comply with the requirements, the batch or blend of batches of composition represented by the sample, shall be rejected. Any squib assemblies that have been loaded with the defective composition shall also be rejected.

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4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8799892 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test method and procedures

4.4.1 Volatile content of flash charge.- Transfer approximately 1.5 grams of the sample, accurately weighed, to a tared, medium porosity sintered glass filtering crucible. Place the crucible and contents in an oven maintained at 60 plus or minus 5 degrees Centrigrade (C.) for two hours, cool in a desiccator and weigh. Percent of total volatiles = $\frac{100A}{W}$

where:

A = loss in weight, grams
W = sample weight, grams

4.4.2 Non-functioning.- The assembly shall be conditioned at a temperature of seventy (70) plus or minus five (5) degrees Fahrenheit (F.) for a min. of two (2) hours prior to testing. After conditioning, the assembly shall be connected to the applicable test equipment and the specified current applied for the required time. This test should be performed behind a suitable barricade.

4.4.3 Functioning.- The following tests shall be performed behind a suitable barricade.

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4.4.3.1 Water immersion.- The assembly shall be immersed to a min. depth of eighteen (18) inches in water maintained at seventy (70) plus or minus five (5) degrees F. for a min. of eight (8) hours. The test circuit shall be prepared for firing by inserting a one ohm precision resistor in the test circuit and adjusting the current through the resistor to five hundred and forty-five (545) plus or minus five (5) milliamperes. After this adjustment of the test circuit the resistor shall be removed. The conditioned squib assembly shall be inserted in the test circuit in the place of the resistor. The circuit shall be energized and the functioning time of the assembly recorded to determine compliance with the requirement. When a retest is performed, the functioning time shall be recorded for information purposes.

4.4.3.2 Cold temperature.- The squib assembly shall be placed in the insulating container (see Note) with approximately four (4) inches of lead wires extending, and the flash charge visible and not more than one half (1/2) inch below the open end of the container. The cover shall be placed on the container and the closed container with squib placed in a conditioning chamber maintained at minus sixty-five (65) plus or minus five (5) degrees F. for a min. of sixteen (16) hours. At the end of the conditioning time, the closed container shall be removed from the cold temperature chamber and the leads connected to the firing circuit (see 4.4.3.1). The cover shall then be removed and the test circuit immediately energized. The functioning time shall be recorded to determine compliance with the requirement. When a retest is performed, the functioning time shall be recorded for information purposes.

NOTE.- The container shall be closed on one end and be manufactured of foamed light weight plastic with approximate dimensions of 1.5 inch inside diameter and 6 inch depth. The cover of the same material shall have sufficient clearance so that the lead wire insulator is not damaged when the cover is installed on the container.

4.4.4 Lead wire pull test.- The lead wires shall be attached to the applicable test fixture and the specified load applied to each lead wire individually along the longitudinal axis of the lead wire. Observation shall be made for evidence of slippage of the lead wire in the holder. Holder assemblies subjected to this test shall not be returned to the lot.

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4.4.5 Lead wire and plug leakage test.- The holder assembly shall be placed in the equipment specified in 4.3.4 and the required vacuum applied. The leakage indicator shall be observed to determine compliance with the requirement.

4.4.6 Bridge wire load test.- The assembly shall be placed in the equipment specified in 4.3.4 and the required load applied on the bridge wire. Observation shall be made for compliance with the requirement. Assemblies subjected to this test shall not be returned to the lot.

4.4.7 Bridge wire resistance test.- The assembly shall be connected to the equipment specified in 4.3.4 and a max. current of ten (10) milliamperes applied. The measuring equipment shall be observed for compliance with the requirement.

4.4.8 Determination of flash charge composition (see 6.6).- The chemical composition of the flash charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate	44.5 plus or minus 2
Lead Thiocyanate	35.5 plus or minus 2
Charcoal	20.0 plus or minus 1.5
Lacquer Solids (added)	1.6 plus or minus 0.2

4.4.8.1 Lacquer solids content.- Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filtering crucible and allowed to air dry for about an hour. The crucible and contents shall be placed in a vacuum oven for two hours maintained at a temperature of 60 plus or minus 5 degrees C. The crucible and contents shall be cooled in a desiccator and weighed. The procedure shall be repeated until the change in weight does not exceed 0.5 milligrams. The content of the crucible shall be extracted with ten 25 milliliter (ml.) portions of acetone saturated with potassium chlorate and lead thiocyanate. The crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and then dried in a vacuum oven maintained at 60 plus or minus 5 degrees C. for one hour. The crucible and contents shall be cooled in a desiccator and weighed. Save the crucible and contents for charcoal analysis. The percent lacquer solids shall be calculated as follows:

$$\text{Percent lacquer solids} = \frac{100A}{B}$$

where:

A = loss in weight, grams
B = weight of residue, grams

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4.4.8.2 Charcoal content.- Extract the contents of the crucible remaining from the lacquer solids determinations with ten 25 ml. portions of hot (90 plus or minus 5 degrees C.) distilled water. Use a small stirring rod to aid in the extraction and apply suction to remove each portion of the extract. Save the filtrate for the determination of lead thiocyanate. Dry the crucible and contents for 2 hours at 100 plus or minus 5 degrees C., cool in a desiccator and weigh.

$$\text{Percent charcoal} = \frac{100A}{B}$$

where:

A = loss in weight, grams
B = weight of residue, grams

4.4.8.3 Lead thiocyanate content.- To the filtrate reserved from the charcoal determination add 35 ml. of standardized 0.1N silver nitrate solution and 100 ml. of 3N nitric acid. The mixture shall be swirled occasionally for five minutes. Add 5 ml. of ferric ammonium sulfate indicator (prepared as specified in MIL-STD-286, Method 705.1) to the solution. Titrate the solution with standardized 0.1N potassium thiocyanate solution to the first permanent reddish brown coloration. The percent lead thiocyanate shall be calculated as follows:

percent lead thiocyanate = 16.17 (AB-CD) divided by E

where:

A = volume of silver nitrate solution used
B = normality of silver nitrate solution
C = volume of potassium thiocyanate solution
D = normality of potassium thiocyanate solution
E = weight of sample after acetone extraction

4.4.8.4 Potassium chlorate content.- The percent potassium chlorate is obtained by subtracting the findings of 4.4.8.2 and 4.4.8.3 from 100 percent.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

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5.1.1 Level C.- The electric squibs shall be packaged in accordance with Code of Federal Regulations 49 CFR Parts 171-179 and the manufacturer's standard practice.

5.2 Packing

5.2.1 Level C.- The electric squibs shall be packed in accordance with Code of Federal Regulations 49 CFR Parts 171-179 and in a manner that will assure safe delivery from the manufacturer's plant to the first receiving activity.

5.3 Marking.- Unit packages shall be identified in accordance with the manufacturer's normal practice. Each exterior shipping container shall be marked in accordance with the requirements of dwg. 8796522. The DOT marking shall be "ELECTRIC SQUIBS".

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and

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calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Duds - A dud is defined as a squib assembly that fails to fire on the first attempt.

6.5 Calculation of sample standard deviation.- The following formula or equivalent shall be used in calculating the sample standard deviation.

$$s = \sqrt{\frac{\Sigma(X-\bar{X})^2}{n-1}}$$

where:

s = sample standard deviation
X = numerical value of individual observations
$\bar{X}$ = average value of the sample
n = sample size

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6.6 Determination of flash charge composition.- During the manufacture of the flash charge composition up to two parts of solvent are added to one part of Egyptian lacquer in order to facilitate the blending of the mixture. Because of the volatility of the solvent, the solvent content of the flash charge composition is found to be a variable. Because of this, the determination of the potassium chlorate, lead thiocyanate and charcoal content are determined on a lacquer free basis, and the lacquer solids on added basis.

6.7 Check test for possible deterioration.- The following test is designed to prevent the loading of squibs which have deteriorated, into a major ammunition item. The test should be performed on squibs from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the squibs are loaded does not require that the check test be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

6.7.1 Water immersion and duds.- One hundred and thirty (130) squibs shall be selected from each lot for this test. The squibs shall be tested as specified in 4.4.3.1. If the mean functioning time plus the product of 2.37 and the standard deviation exceeds .10 second, the lot shall be rejected. Duds shall not enter into the calculation for standard deviation. If during the performance of this test two or more are encountered the lot shall be rejected. If only one (1) dud is encountered, a retest shall be permitted using double (260) the number of samples used in the original test. If during the retest one more dud is encountered, testing shall be stopped and the lot rejected.

6.7.2 Non-functioning.- A sample of 32 squibs shall be selected from each lot for this test. The test shall be performed as specified in 4.4.2 using test equipment specified in 4.3.4. If one or more squibs fail to comply with the specified requirement the lot shall be rejected.

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1375-A-042

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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e to this specification. The examined visually to determine conformanc

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DOVER, NEW JERSEY 07801

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MILITARY SPECIFICATION

SQUIB, ELECTRIC, M2
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-S-45428D(MU), dated 7 May 1969, and is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.6 Workmanship. Add the following to end of this paragraph:

"The soldered joints shall be flushed of all residual flux by means of a cleaning solvent approved by the Contracting Officer. There shall be no chlorides present at the soldered joints when tested in accordance with 4.4.8.5.

PAGE 8

4.3.2.5 Squib assembly. In table for Major Defects add the following under the headings shown below, as they appear on Page 8:

Categories	Defects	Method of Inspection	Code No.
Major			
"105	Presence of chloride at soldered joints	See 4.3.3.9	05007"

PAGE 10

Add an additional paragraph, and number it 4.3.3.9:

"4.3.3.9 Presence of chloride at soldered joints (see dwg. 8799892 and 4.3.2.5 category 105 - major defect, Code No. 05007).

FSC 1375

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The sampling plan for this test shall be in accordance with MIL-STD-105 and 100 percent inspection. The lot shall be rejected if it fails to comply with this sampling plan. The test shall be performed as specified in 4.4.8.5."

PAGE 14

Add an additional paragraph, and number it 4.4.8.5:

"4.4.8.5 Chloride contamination test. CAUTION - Proper laboratory procedures and practices must always be followed. Since nitric acid and silver nitrate are corrosive to skin, the use of eye and face protection as well as rubber gloves should be mandatory. Additional safety rules of individual laboratories or operating areas should be observed."

4.4.8.5.1 Material and equipment.

a. Assorted laboratory beakers.
b. Deionized and/or distilled water.
c. ACS reagent grade isopropyl alcohol.
d. Nitric acid (70%, for trace metal analysis).
e. Silver nitrate solution (0.1N).
f. Nitric acid solution (35%). Prepare a diluted nitric acid solution by carefully adding a volume of the 70% nitric acid to an equal volume of water. For safety reasons, add the acid to the water. NOTE: The acid may be stored in a proper bottle for later use. The acid solution should be at room temperature before use.

4.4.8.5.2 Additional instructions.

a. Ensure that beakers as well as the water are free of chloride ion contamination prior to use. Laboratory apparatus should be pre-rinsed with distilled or deionized water prior to use.

b. Only deionized or distilled water shall be used.

c. Beaker and water used should be tested by adding a few drops of diluted nitric acid and then a few drops of silver nitrate solution and mix. The presence of any chlorides or other halides will cloud the solution..

4.4.8.5.3 Test procedure. Take 200 plug assemblies and rinse the bridge portion of the plug assembly in a beaker containing 100ml of rinse solution. This solution shall consist of 75% by

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volume of isopropyl alcohol and 25% deionized and/or distilled water. Add a few drops of diluted nitric acid and then a few drops of silver nitrate solution to the rinse solution. Mix the solution by swirling the beaker. Any milky cloudiness of the solution indicates the presence of halide (Chlorides, bromide or iodide) ions."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A250)

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NOTICE
OF VALIDATION

MIL-S-45428D(AR)
NOTICE 1
16 DECEMBER 1989

MILITARY SPECIFICATION

SQUIBB, ELECTRIC, M2 PARTS FOR
AND LOADING, ASSEMBLING AND PACKING

MIL-S-45428D(AR), dated 7 May 1969, with Amendment 2, dated 1 November 1982, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Army - AR

AMSC N/A

FSC 1375

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